

Alejandro García Castellanos

Ph.D. Candidate Machine Learning
AMLab, University of Amsterdam

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RESEARCH INTERESTS

Applying topology, geometry, and algebra to the representation learning and generative modeling of shapes for enhanced human-AI collaboration.

EXPERIENCE

Ph.D. Candidate

Feb. 2024 – Present

AMLab (Amsterdam Machine Learning Lab), UvA

Amsterdam, Netherlands

- Under the supervision of Erik Bekkers (UvA) and co-supervision of Daniël Pelt (ULEI)
- Sponsored by [The Hybrid Intelligence Centre](#)
- Geometry-informed machine learning techniques for collaborative human-computer image annotation

Research Engineer

Mar. 2023 – Feb. 2024

RPL (Division of Robotics, Perception and Learning), KTH

Stockholm, Sweden

- Under the supervision of Danica Kragic
- Metric learning in generative models
- Hierarchy discovery in Hyperbolic geometry

Trainee Internship

Sep. 2020 – Jan. 2021

MEDAL laboratory, CTB (Center of Biomedical Technology)

Madrid, Spain

- Under the supervision of Ernestina Menasalvas
- Development of new techniques for information recognition in medical corpus
- Fine-tuning an LLM in breast and lung cancer's clinical notes

EDUCATION

MSc in Machine Learning

Sep. 2021 – Oct. 2023

KTH Royal Institute of Technology

Stockholm, Sweden

- Average grade: 4.00 GPA
- Applied Mathematics elective track: Topological Data Analysis, Regression Analysis, Time Series Analysis
- Thesis in “*Topological regularization and relative latent representations*”

B.S. in Mathematics and Computer Science

Sep. 2017 – Jul. 2021

Universidad Politécnica de Madrid

Madrid, Spain

- Average grade: 9.78/10.0 (4.00 GPA)
- Best academic record amongst all programs at the ETSIINF (Higher Technical School of Computer Engineers) of the promotion 2020/2021
- Honours granted by UPM in 29 courses
- Three Scholarships for Academic Excellence granted by the Community of Madrid
- Thesis in “*Robustness of Persistent Homology: the Stability Theorem*”

SKILLS

Programming: Python, Java, C, R, SQL, MATLAB, Maple, Prolog, Bash

ML Frameworks: PyTorch, PyTorch Lightning, JAX, TensorFlow

Tools: $\text{\LaTeX}$, Git, Microsoft Office, Keynote, Pages, Adobe Photoshop, Adobe Lightroom

Soft Skills: public speaking, written communication, teamwork, project and time management, critical thinking, analytical skills, research skills, creativity, strategic thinking

Reinforcement Learning

2025, 2026

Teaching assistant

Master In Artificial Intelligence, UvA

Deep Learning 2

2024, 2025

Teaching assistant and project supervisor

Master In Artificial Intelligence, UvA

Master Thesis Supervisor

2025

Thesis: “Persistent Homology Go Combinatorial”

Master In Artificial Intelligence, UvA

RESEARCH PUBLICATIONS

(* := EQUAL CONTRIBUTION)

- AIStats '26 **A. García-Castellanos***, A. A. Medbouhi*, G. L. Marchetti, D. M. Pelt, E. J. Bekkers, D. Kragic (2025). “*Randomized HyperSteiner: A Stochastic Delaunay Triangulation Heuristic for the Hyperbolic Steiner Minimal Tree*”.
- NeurIPS '25 **A. García-Castellanos**, D. R. Wessels, N. J. Berg, R. Duits, D. M. Pelt, E. J. Bekkers (2025). “*Equivariant Eikonal Neural Networks: Grid-Free, Scalable Travel-Time Prediction on Homogeneous Spaces*”. In: *Advances in Neural Information Processing Systems (NeurIPS)*.
- ALENEX '25 **A. García-Castellanos***, A. A. Medbouhi*, G. L. Marchetti, E. J. Bekkers, D. Kragic (2025). “*Hypersteiner: Computing heuristic hyperbolic steiner minimal trees*”. In: *2025 Proceedings of the Symposium on Algorithm Engineering and Experiments (ALENEX)*.
- TAG-DS '25 O. Boufalis*, J. Carrasco-Pollo*, J. Rosenthal*, E. Terres-Caballero*, **A. García-Castellanos** (2025). “*Symmetry-Aware Graph Metanetwork Autoencoders: Model Merging through Parameter Canonicalization*”. In: *Proceedings of the 1st Conference on Topology, Algebra, and Geometry in Data Science (TAG-DS)*. **Oral presentation**.
- UniReps '24 **A. García-Castellanos**, G. L. Marchetti, D. Kragic, M. Scolamiero (2024). “*Relative representations: Topological and geometric perspectives*”. In: *Proceedings of the II edition of the Workshop on Unifying Representations in Neural Models (UniReps)*.

PREPRINTS & MANUSCRIPTS

(* := EQUAL CONTRIBUTION)

- Preprint
(Ext. Abs.) **A. García-Castellanos**, M. Weiler, E. J. Bekkers (2026). “*RoVE: Rotary Value Embeddings Attention for Relative Position-dependent Value Pathways*”.
- GRaM '26
(Ext. Abs.) **A. García-Castellanos**, G. Bellaard, R. Duits, D. M. Pelt, E. J. Bekkers (2026). “*Generalized Reduction to the Isotropy for Flexible Equivariant Neural Fields*”. In: *Extended Abstract track of the ICLR 2026 Workshop on Geometry-grounded Representation Learning and Generative Modeling (GRaM)*.
- UniReps '25
(Ext. Abs.) B. Kuipers*, F. Byrman*, D. Uytterlinde* **A. García-Castellanos** (2025). “*Symmetry-Aware Fully-Amortized Optimization with Scale Equivariant Graph Metanetworks*”. In: *Extended Abstract track of the III edition of the Workshop on Unifying Representations in Neural Models (UniReps)*.
- GRaM '24
(Blog post) D. Canez*, N. Midavaine*, T. Stessen*, J. Fan*, S. Arias*, **A. García-Castellanos** (2024). “*Effect of equivariance on training dynamics*”. In: *Blog post track of the I edition of the Workshop on Geometry-grounded Representation Learning and Generative Modeling (GRaM)*.

GRaM '24
(Blog post) M. Carrasco*, A. Berentzen*, **A. García-Castellanos** (2024). “*Applications of TopoX to Equivariant Topological Networks*”. In: *Blog post track of the I edition of the Workshop on Geometry-grounded Representation Learning and Generative Modeling (GRaM)*.

INVITED TALKS

4th workshop on Mathematics and AI <i>Tilburg, Netherlands</i>	June 2025
Learning on Graphs Conference <i>Amsterdam, Netherlands</i>	November 2024

COMMUNITY SERVICE

Reviewer <i>NeurIPS Proceedings</i>	2025
Reviewer <i>NeurReps Workshop</i>	2025
Reviewer <i>UniReps Workshop</i>	2024
Reviewer <i>ICML Proceedings</i>	2024

LANGUAGES

English: Advanced level (C1), CAE by Cambridge English

Spanish: Native language

German: Low Beginner Level